MATH1101: Differential Calculus & Analytical Geometry

Complete Exam Notes · 1st Year 1st Semester · PUST · Module 7 (Units 19, 20 & 21): 3D Coordinates, DCs/DRs, Projections, Planes & Straight Lines

Bismillahir Rahmanir Rahim · Md. Ataullah | Roll: 260503 | Reg: 1054600 | Dept. of EECE, PUST (18th Batch, 2025–26)

Module 7 Overview

Units Covered in Module 7:

- **Unit 19:** Three-Dimensional Coordinate Systems (Cartesian, Cylindrical & Spherical Coordinates).
- **Unit 20:** Direction Cosines (DCs), Direction Ratios (DRs), Projections, and Angle Between Two Lines.
- **Unit 21:** Planes, Straight Lines in 3D, Coplanarity, and Shortest Distance (S.D.) Between Two Skew Lines.

Master Roadmap (MATH 1101)

Mod.	Units & Core Syllabus Topics	Main Sources	Status
1	Units 1–3: Functions, Domain/Range, Graphs, Inverse, Limits & Continuity	Class Notes · Anton Ch. 0, 1 · PUST 20–23	Module 1
2	Units 4–5: Differentiability, Geometric Meaning, Extrema, y_n & Leibniz	Class Notes · Anton Ch. 2, 4 · PUST 20–23	Module 2
3	Units 6–9: Rolle's & MVT, Taylor/Maclaurin, L'Hôpital & Curve Tracing	Class Notes · Anton Ch. 3, 4, 9 · Handout	Module 3
4	Units 10–13: Tangents, Asymptotes, Curvature, Partial Deriv., Euler & Jacobian	Akib Notes · Anton Ch. 13 · PUST 20–23	Module 4
5	Units 14–15: 2D Coordinates, Transformation of Axes & Pair of Straight Lines	Anton Ch. 10 · Rahman & Bhattacharjee	Module 5
6	Units 16–18: Conic Reduction, Parabola, Ellipse, Hyperbola & System of Circles	Anton Ch. 10 · Rahman & Bhattacharjee	Module 6
7	Units 19–21: 3D Coordinates, DCs/DRs, Projections, Planes & Skew Lines (S.D.)	Anton Ch. 11 · Rahman & Bhattacharjee	✓ **This file**
8	Units 22–24: Sphere, Cone, Right Circular Cylinder & 3D Central Conicoids	Anton Ch. 11 · Rahman & Bhattacharjee	Module 8

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Conversion	Forward Equations	Inverse Equations & Restrictions
Cylindrical $(r,\theta,z)\leftrightarrow$ **Cartesian** (x,y,z)	$x=r\cos\theta,\quad y=r\sin\theta,\quad z=z$	$r^2=x^2+y^2,\quad \tan\theta=\dfrac{y}{x},\quad z=z$ $(r\geq 0,\ 0\leq\theta<2\pi)$.
Spherical $(\rho,\phi,\theta)\leftrightarrow$ **Cartesian** (x,y,z)	$x=\rho\sin\phi\cos\theta, y=\rho\sin\phi\sin\theta, z=\rho\cos\phi$	$\rho^2=x^2+y^2+z^2,\ \cos\phi=\dfrac{z}{\rho},\quad \tan\theta=\dfrac{y}{x}$ $(\rho\geq 0,\ 0\leq\phi\leq\pi,\ 0\leq\theta<2\pi)$.
Spherical $(\rho,\phi,\theta)\leftrightarrow$ **Cylindrical** (r,θ,z)	$r=\rho\sin\phi,\quad \theta=\theta,\quad z=\rho\cos\phi$	$\rho=\sqrt{r^2+z^2},\quad \tan\phi=\dfrac{r}{z},\quad \theta=\theta.$

1 Unit 19: 3D Coordinate Systems (Cartesian, Cylindrical & Spherical)

2 Unit 20: Direction Cosines, Direction Ratios, Projections & Angle Between Lines

2.1 Definitions and Core Formulas

1. **Direction Cosines (l,m,n):** If a directed line makes angles α,β,γ with the positive x-, y-, and z-axes, then $l=\cos\alpha$, $m=\cos\beta$, $n=\cos\gamma$ satisfy:

$$l^2+m^2+n^2=\cos^2\alpha+\cos^2\beta+\cos^2\gamma=1.$$

2. **Direction Ratios (a,b,c):** Numbers proportional to (l,m,n):

$$l=\frac{\pm a}{\sqrt{a^2+b^2+c^2}},\qquad m=\frac{\pm b}{\sqrt{a^2+b^2+c^2}},\qquad n=\frac{\pm c}{\sqrt{a^2+b^2+c^2}}.$$

3. **Projection of Segment PQ Joining $P(x_1,y_1,z_1)$ and $Q(x_2,y_2,z_2)$ on a Line with DCs (l,m,n):**

$$\text{Projection}=l(x_2-x_1)+m(y_2-y_1)+n(z_2-z_1).$$

4. **Angle θ Between Two Lines:**

$$\cos\theta=l_1l_2+m_1m_2+n_1n_2=\frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}.$$

2.2 Solved PUST Past Exam Problems on Direction Cosines

PUST Past Exam Question 1 (Cube Diagonal Theorems) [5 Marks]
Question: (a) Find the angle between any two diagonals of a cube. (b) Prove that if a line makes angles $\alpha,\beta,\gamma,\delta$ with the four diagonals of a cube, then $\cos^2\alpha+\cos^2\beta+\cos^2\gamma+\cos^2\delta=\dfrac{4}{3}$.

Solution (Exam Style): Let $O(0,0,0)$ be one corner of a cube of side a with edges along the axes. The direction cosines of the four diagonals OP, AL, BM, CN are:

$$\left(\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right),\quad \left(-\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right),\quad \left(\frac{1}{\sqrt{3}},-\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right),\quad \left(\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},-\frac{1}{\sqrt{3}}\right).$$

Part (a):

$$\cos\theta=\left|-\frac{1}{3}+\frac{1}{3}+\frac{1}{3}\right|=\frac{1}{3}$$

$$\implies \theta = \cos^{-1}\left(\frac{1}{3}\right).$$

Part (b): If (l, m, n) are the direction cosines of the line (with $l^2 + m^2 + n^2 = 1$):

$$\cos\alpha = \frac{l+m+n}{\sqrt{3}}, \qquad \cos\beta = \frac{-l+m+n}{\sqrt{3}},$$
$$\cos\gamma = \frac{l-m+n}{\sqrt{3}}, \qquad \cos\delta = \frac{l+m-n}{\sqrt{3}}$$
$$\implies \cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta = \frac{4(l^2+m^2+n^2)}{3} \quad \text{[Cross-product terms cancel in pairs]}$$
$$= \frac{4(1)}{3} = \frac{4}{3}. \quad \blacksquare$$

PUST Past Exam Question 2 (Algebraic DC Relations) [5 Marks]

Question: Find the angle between the two lines whose direction cosines (l, m, n) satisfy $l + m + n = 0$ and $l^2 + m^2 - n^2 = 0$.

Solution (Exam Style):

$l + m + n = 0 \implies n = -(l + m)$

Substituting into $l^2 + m^2 - n^2 = 0$: $\quad l^2 + m^2 - (l+m)^2 = 0$

$$\implies -2lm = 0$$
$$\implies l = 0 \quad \text{or} \quad m = 0.$$

- **When $l = 0$:** $n = -m \implies$ DRs of Line 1: $(0, m, -m) \propto (0, 1, -1)$.
- **When $m = 0$:** $n = -l \implies$ DRs of Line 2: $(l, 0, -l) \propto (1, 0, -1)$.

$$\therefore \cos\theta = \frac{(0)(1) + (1)(0) + (-1)(-1)}{\sqrt{0^2+1^2+(-1)^2}\sqrt{1^2+0^2+(-1)^2}} = \frac{1}{2}$$
$$\implies \theta = 60^\circ = \frac{\pi}{3}.$$

3 Unit 21: Planes, Straight Lines in 3D & Shortest Distance

3.1 Equations of a Plane

1. **General Equation:** $Ax + By + Cz + D = 0$ (normal DRs: A, B, C).
2. **Intercept Form:** $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.
3. **Plane Through Intersection of $P_1 = 0$ and $P_2 = 0$:** $P_1 + \lambda P_2 = 0$.
4. **Perpendicular Distance from (x_1, y_1, z_1) to $Ax + By + Cz + D = 0$:** $d = \frac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2 + B^2 + C^2}}$.

PUST Past Exam Questions on Planes [5 Marks Each]

Problem A: Find the plane through the intersection of $x+2y+3z-4=0$ and $2x+y-z+5=0$ perpendicular to $5x+3y+6z+8=0$.

Solution (Exam Style):

Plane family: $(x+2y+3z-4)+\lambda(2x+y-z+5)=0$

$\Longrightarrow (1+2\lambda)x+(2+\lambda)y+(3-\lambda)z+(5\lambda-4)=0$

Perpendicularity condition: $5(1+2\lambda)+3(2+\lambda)+6(3-\lambda)=0$

$\Longrightarrow 7\lambda+29=0 \Longrightarrow \lambda=-\dfrac{29}{7}$

$\Longrightarrow 7(x+2y+3z-4)-29(2x+y-z+5)=0$

$\Longrightarrow 51x+15y-50z+173=0.$

Problem B: A variable plane at constant distance p from the origin meets the axes at A,B,C. Show that the locus of the centroid of $\triangle ABC$ is $x^{-2}+y^{-2}+z^{-2}=9p^{-2}$.

Solution (Exam Style):

Let the plane be $\dfrac{x}{a}+\dfrac{y}{b}+\dfrac{z}{c}=1$ [Meets axes at $A(a,0,0)$, $B(0,b,0)$, $C(0,0,c)$]

Centroid $(\alpha,\beta,\gamma)=\left(\dfrac{a}{3},\dfrac{b}{3},\dfrac{c}{3}\right) \Longrightarrow a=3\alpha,\quad b=3\beta,\quad c=3\gamma$

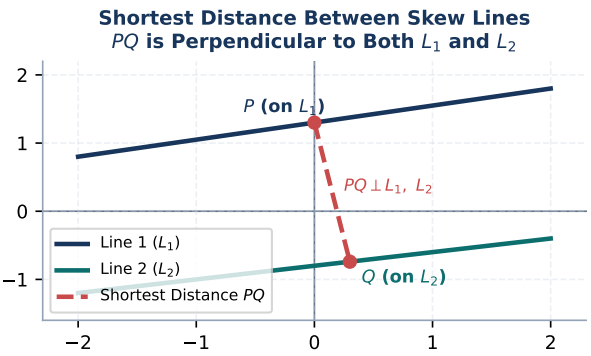
Distance from origin $p=\dfrac{1}{\sqrt{a^{-2}+b^{-2}+c^{-2}}} \Longrightarrow \dfrac{1}{a^2}+\dfrac{1}{b^2}+\dfrac{1}{c^2}=\dfrac{1}{p^2}$

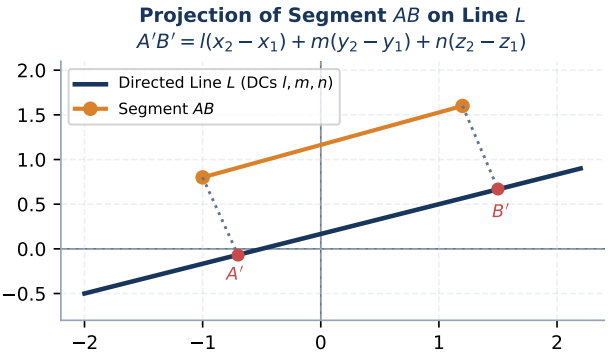
$\Longrightarrow \dfrac{1}{9\alpha^2}+\dfrac{1}{9\beta^2}+\dfrac{1}{9\gamma^2}=\dfrac{1}{p^2} \Longrightarrow x^{-2}+y^{-2}+z^{-2}=9p^{-2}.$ ■

3.2 The Straight Line in 3D and Shortest Distance (S.D.) Between Skew Lines

1. **Symmetric Form:** $\dfrac{x-x_1}{l}=\dfrac{y-y_1}{m}=\dfrac{z-z_1}{n}=r \Longrightarrow$ general point $(x_1+lr,\ y_1+mr,\ z_1+nr)$.
2. **Coplanarity Condition for Two Lines:**

$$\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix}=0.$$





PUST Past Exam Question (Shortest Distance & Equations of S.D. Line) [7 Marks]

Question: Find the length and the equations of the line of shortest distance between the two skew lines:

$$\frac{x-3}{3}=\frac{y-8}{-1}=\frac{z-3}{1} \quad \text{and} \quad \frac{x+3}{-3}=\frac{y+7}{2}=\frac{z-6}{4}.$$

Solution (Exam Style):

Let $P(3r_1+3,\ -r_1+8,\ r_1+3)$ be on Line 1, and $Q(-3r_2-3,\ 2r_2-7,\ 4r_2+6)$ on Line 2.
Direction ratios of PQ : $(-3r_2-3r_1-6,\ \ 2r_2+r_1-15,\ \ 4r_2-r_1+3)$.

Since PQ is perpendicular to both Line 1 (DRs: $3,-1,1$) and Line 2 (DRs: $-3,2,4$):

$$3(-3r_2-3r_1-6)-1(2r_2+r_1-15)+1(4r_2-r_1+3)=0 \implies -11r_1-7r_2=0,$$
$$-3(-3r_2-3r_1-6)+2(2r_2+r_1-15)+4(4r_2-r_1+3)=0 \implies 7r_1+29r_2=0$$
$$\implies r_1=0, \qquad r_2=0.$$

Substituting $r_1=0, r_2=0$ gives the feet of the common perpendicular:

$P=(3,8,3), \qquad Q=(-3,-7,6), \qquad$ DRs of $PQ=(-6,-15,3)\propto(2,5,-1)$
Length of S.D. $=|PQ|=\sqrt{(-3-3)^2+(-7-8)^2+(6-3)^2}=\sqrt{36+225+9}=3\sqrt{30}$
Equation of S.D. Line: $\dfrac{x-3}{2}=\dfrac{y-8}{5}=\dfrac{z-3}{-1}.$

Quick Revision — Module 7 (Units 19–21)

- **DCs & Projections:** $l^2+m^2+n^2=1$; $\cos\theta=l_1l_2+m_1m_2+n_1n_2$; Cube diagonals: $\theta=\cos^{-1}(1/3)$, $\sum\cos^2\alpha=4/3$.
- **Planes:** Intersection family: $P_1+\lambda P_2=0$; Distance: $d=\frac{|Ax_1+By_1+Cz_1+D|}{\sqrt{A^2+B^2+C^2}}$.
- **Shortest Distance (S.D.):** Take general points $P(r_1)$ on L_1 and $Q(r_2)$ on L_2, enforce $PQ\perp L_1$ and $PQ\perp L_2$ to find r_1,r_2, then S.D. $=|PQ|$.

Common Mistakes to Avoid in the Exam

- Using Direction Ratios (a,b,c) instead of normalized Direction Cosines (l,m,n) in the projection formula Proj $=l(x_2-x_1)+m(y_2-y_1)+n(z_2-z_1)$.